

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION II
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/03

Paper 3 Applications Paper

25 September 2013

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

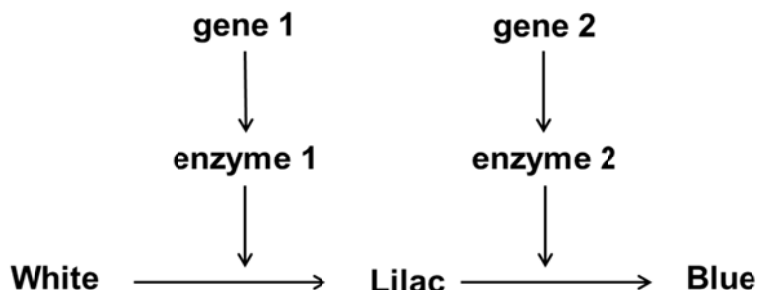
For Examiner's Use	
1	15
2	15
3	10
4	12
5	20
Total	72
Percentage	
Grade	

This document consists of **16** printed pages.



Section AAnswer **all** questions.

- 1 In flax plants, the flowers are white, lilac or blue. Fig.1 shows the pathway by which the flower cells produce coloured pigments.

**Fig. 1**

- (a) A single base deletion occurs in gene 1.

Describe and explain the effect of this mutation on the colour of flax plant flowers.

1. *frameshift mutation/ change in reading frame of gene/ change in codons downstream of mutation;;*
2. *change in aa seq. of enz 1 → change in R groups → change in R group interactions in polypeptide;;*
3. *leading to different folding thus 3D configuration of enz 1 → likely to be non-functional / does not catalyze synthesis of lilac pigment;;*
4. *lack of substrate for enz 2 to act on to synthesize blue pigment → flax flowers appear white;;*

[4]

- (b) Gel electrophoresis was used to separate the enzymes involved in this pathway. When extracts of the differently coloured flax petals were analyzed, four different patterns of bands were produced. In Table 1, only bands that contain functional enzymes are shown.

Table 1

sis	s
	
	<i>blue</i>
	<i>lilac</i>
	<i>white</i>

- (i) With reference to Fig. 1, complete Table 1 to give the colour of the petal from which each extract was taken. [3]
- (ii) Describe how the enzymes were separated by gel electrophoresis.

1. *polyacrylamide gel electrophoresis for separation of proteins;;*

2. *SDS to denature / linearize prot to remove influence of shape on migration speed and confer equal negative charge per unit mass/ length using sodium dodecyl sulfate (SDS) / anionic detergent;;*

3. *Buffer used to provide medium for ion conductivity;;*

4. *load prot sample in wells near the negative electrode / cathode → apply voltage / electric field/ direct current across gel;;*

5. *-vely charged prot migrate towards anode, polyacrylamide gel acts as molecular sieve to impede movement of larger prot/ larger prot travels shorter distance compared to smaller prot;;* [5]

- (c) Genomic libraries and cDNA libraries can be created for the flax plant genome.

State **three** differences between the two gene libraries.

Feature	Genomic library	cDNA library
1. Content	Contains entire <u>DNA content</u> of an organism including all coding and non-coding sequences/ introns + exons	Contains entire <u>protein-encoding</u> DNA content/ exons only
2. Starting genetic material to be isolated	Requires <u>chromosomal/ total genomic DNA</u> isolation	Requires <u>total mRNA isolation</u> → reverse transcribed into cDNA;
3. Starting cellular material	Can be from <u>any cell/ tissue</u>	Total mRNA should be isolated from a <u>specific cell/ tissue</u> where the particular protein of interest is likely to be produced in large quantities
4. Intactness of genes.	A gene may be cut in between when there is a restriction site within the sequence	<u>Intact</u> genes are obtained as cDNAs are derived from mRNAs
5. Size of library	<u>Larger</u> number of different clone types in the library as fragments originate from whole genome	<u>Smaller</u> number of different clone types in the library (less than 1% of genome)
6. Functional use of library in relation to type of sequences studied.	use to study gene expression due to presence of reg seq like promoters	usually to decipher aa seq / clone to produce polypeptide
7. Functional use of library in relation to patterns of gene expression studied.	<u>Cannot</u> be used to study physiological or developmental-based changes in gene expression	<u>Can</u> be used for tracing changes in patterns of gene expression under different developmental or physiological conditions

(Any 3, 1 mark each)

[3]

[Total: 15]

- 2 Diabetes mellitus is a human disease condition in which the patients do not produce sufficient insulin. A way to treat diabetic patients is to inject purified recombinant human insulin, produced using genetically engineered bacteria, into their bloodstream.

Insulin is a relatively small protein comprising two polypeptides, one of 21 amino acids (the A chain) and one of 30 amino acids (the B chain). Trinucleotides representing all the codons are synthesized and joined together to form two artificial genes for the respective chains. Fig. 2.1 shows the two artificial genes, including the positions of the recognition sequences of two restriction enzymes, *Pst*I and *Eco*RI, and distances (in base pairs) between the sites.

Each artificial gene is cloned into a separate bacterial plasmid pUC21, using the restriction enzyme *Hind*III, and placed under the control of a strong *lac* promoter. Fig. 2.2 shows the plasmid map of pUC21. It includes the positions of the restriction sites for the three restriction enzymes, *Pst*I, *Eco*RI and *Hind*III, and distances (in base pairs) between the sites. The shaded regions represent two marker genes, *Amp*^R and *lac* Z.

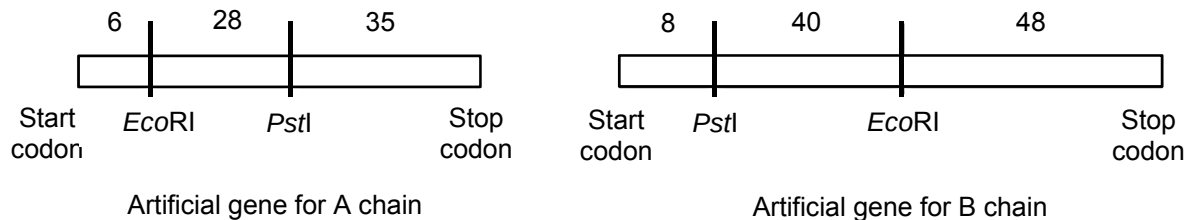


Fig. 2.1

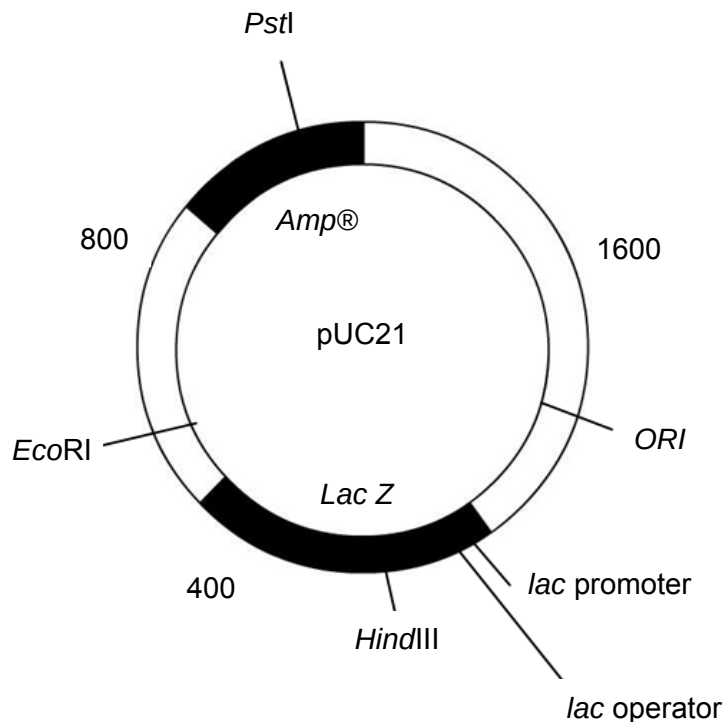


Fig. 2.2

Both recombinant plasmids containing the two artificial genes are transformed separately into *E. coli* cells. The bacterial cells from the transformation process are grown on an agar plate containing ampicillin and X-Gal.

- (a) State the substance that must be added to induce the expression of the artificial genes in *E. coli*.

Lactose/ allolactose/ IPTG;;

[1]

- (b) Explain what is meant by a *marker gene*.

Gene which will give a visible/ detectable phenotype / trait /characteristic to the bacteria for identification/ selection of bacteria containing recombinant plasmid/ gene of interest;;

[1]

- (c) Heat shock treatment is applied to *E.coli* cells to render them competent to take up recombinant plasmids.

Describe **three** ways in which this process of artificial transformation of the recombinant plasmid into bacterial cells differs from natural transformation.

<i>Artificial Transformation</i>	<i>Natural Transformation</i>
<i>1. plasmids enter via pores induced from heat shock</i>	<i>1. plasmids enter by binding to competence factor / cell surface receptor of bacterial cell</i>
<i>2. dsDNA enters</i>	<i>2. ssDNA enters</i>
<i>3. does not undergo homologous recombination, exist as plasmid/ extrachromosomal DNA</i>	<i>3. allele may undergo recombination with homologous region in host chrom</i>

[3]

- (d) Describe and explain the procedure to select for bacterial cells containing the recombinant plasmid.

- 1. Bacterial cells successfully transformed with plasmids will be resistant to ampicillin hence can survive/ grow/ form colonies;;***
- 2. bacterial cells containing non-recombinant plasmids appear blue since lacZ gene is intact/ not disrupted by insertion of insulin genes → functional β-galactosidase produced***
- 3. bacterial cells containing recombinant plasmids appear white since lacZ gene has been disrupted/ insertional inactivation of lacZ gene has occurred → no functional β-galactosidase produced***
- 4. → X-gal hydrolyzed/ cleaved/ → X-gal not hydrolyzed/ cleaved;; (only awarded if pt2/3 is correct)***

[4]

Once the transformed bacterial colonies have been identified, nucleic acid hybridization is carried out to further identify the colonies which contained the artificial genes.

The plasmid DNA from these colonies is then isolated and subjected to digestion by a particular restriction enzyme to determine which colonies have the artificial genes inserted in the correct orientation.

A student who is unsure of which particular restriction enzyme to use carried out digestion of all the four possible recombinant plasmids numbered A1, A2, B1 and B2 using either enzyme *EcoRI* or *PstI*.

Table 2 shows all the results obtained.

Table 2

	Digestion of plasmid DNA with artificial gene for A chain				Digestion of plasmid DNA with artificial gene for B chain			
Plasmid	A1	A2	A1	A2	B1	B2	B1	B2
Length of fragments (bp)	406	463	1234	1235	1288	1208	448	448
	2463	2406	1635	1634	1608	1688	2448	2448
Restriction enzyme used	<i>EcoRI</i> ;		<i>PstI</i> ;		<i>PstI</i> ;		<i>EcoRI</i> ;	

- (e) With reference to Fig. 2.1 and Fig. 2.2, complete Table 2 by identifying the particular restriction enzyme used for digestion of the respective plasmid DNA. [2]
- (f) With reference to Fig. 2.1, Fig. 2.2 and Table 2, state the appropriate restriction enzyme that has to be used for the digestion of the respective plasmid DNA in order to determine which has the artificial genes inserted in the correct orientation.
- (i) Restriction enzyme to digest plasmid DNA with artificial gene *EcoRI*;;
for A chain
- (ii) Restriction enzyme to digest plasmid DNA with artificial gene *PstI*;;
for B chain

[2]

(g) Provide an explanation for your answer in (f).

1. *artificial genes must be cut in such a manner that the fragments generated are of different lengths / not of the same length;;*

2. *EcoRI cleavage generates 406kb fragment for chain A/1208kb fragment for chain B if wrong orientation indicating start codon attached @ a further site from promoter (accept reverse aug)* [2]

[Total: 15]

3 Duchenne muscular dystrophy (DMD) is a common lethal X-linked human genetic disease caused by the absence of the protein dystrophin in muscle fibres.

Possible treatments included the transplanting of normal cells from genetically compatible donors or gene therapy which involves incorporating recombinant DNA with part of the normal allele of the gene for dystrophin into a muscle.

(a) Explain briefly what is meant by *genetically compatible donors*.

1. *individuals who are genetically similar to recipient / blood relations/ close relatives;;*

2. *transplanted cells from donor would not elicit cell rejection / trigger immune response;;*

3. *antigens on CSM are not recognized as foreign / donor cells with similar glycoprot on CSM;;* [2]

The dystrophin gene was inserted into an adenovirus and injected into a strain of mice with the symptoms of muscular dystrophy.

Eight mice with DMD were injected once in a muscle in one hind limb with different doses of the modified virus 7 days after birth. The mice were then killed at the age of 2 to 25 weeks and sections of the injected muscle and uninjected muscle from the other hind limb were obtained.

These tissues were then treated with a fluorescent antibody specific to dystrophin to show gene expression and the results are shown in Table 3.

Table 3

Mouse	Virus dose/ particles*10 ⁹ cm ⁻³	Death/weeks after birth	Increase in muscles fibres with dystrophin/%
1	1.0	2	6
2	2.0	3	8
3	2.5	5	10
4	3.0	8	22
5	4.0	10	30
6	5.0	14	50
7	5.0	18	35
8	5.0	25	15

- (b) Describe and explain how the adenovirus delivers the dystrophin gene.
1. *adenovirus binds to specific receptors found on CSM of muscle cells, enters by receptor-mediated endocytosis;;*
 2. *viral capsid enters cytoplasm, enzymes remove capsid to release viral genome along with normal dystrophin gene;;*
 3. *normal dystrophin gene travels into nucleus of muscle cell, undergoes homologous recombination with host chromosome to replace defective allele / exist and expressed extra-chromosomally to express dystrophin prot;;*
- [3]

(c) With reference to Table 3,

- (i) suggest why uninjected muscle from each mouse was also treated with fluorescent antibody specific to dystrophin.

act control

to calculate % increase in muscle fibres w dystrophin by subtracting away baseline production from uninjected muscle;;

[1]

- (ii) describe the effect of the injected dosage of virus on the percentage of muscle fibres producing dystrophin.

1. *higher dosage of virus, higher the increase in % of muscle fibre producing dystrophin*

$1.0 \times 10^9 \text{ cm}^{-3}$ with 6% increase at 2 weeks to $5.0 \times 10^9 \text{ cm}^{-3}$ with 50% increase at 14 weeks;;

2. *virus dose of $5.0 \times 10^9 \text{ cm}^{-3}$ for mice killed in 14 – 25 weeks caused decreased in % increase of muscle fibre producing dystrophin*

from 50% to 15%;;

[2]

- (iii) explain why this gene therapy needs to be repeated.

1. *transient expression / short term effect;*

2. *at high dose of virus @ $5.0 \times 10^9 \text{ cm}^{-3}$, % increase drops from 50% for mice in week 14 to 15% in week 25;*

3. *normal dystrophin gene did not integrate into host chromosome;*

4. *not segregated equally into daughter cells during mitosis / lost when the cell undergoes replicative senescence / apoptosis;;*

[2]

(any 2 points to score 1m)

[Total: 10]

4 Planning question

Malt is germinated cereal grains which have been dried in a process known as malting. Malted barley contain amylases which digest starch into maltose units by cleaving the α -1,4-glycosidic bonds.

The digestion of starch is an exothermic process and releases energy. This energy released can be detected by the largest increase in temperature of the reaction mixture over a period of time.

Using this information and your own knowledge, design an experiment to determine the concentration of active amylase in malted barley samples that have undergone different heat treatments.

You must use:

- 1.0 moldm⁻³ starch solution
- 10% amylase solution
- Malted barley which have heat treated for 10 minutes at 40.0 °C, labelled **M1**
- Malted barley which have been heat treated for 10 minutes at 80.0°C, labelled **M2**
- distilled water
- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes, pipette fillers
- timer e.g. stopwatch or stopclock
- mercury thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms, include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Theoretical consideration

1. [mode of enzyme action = 1 mark]

- Starch is a polysaccharide consisting α -glucose molecules linked by α -1,4 and α -1,6 glycosidic bonds.
- Amylase is an enzyme which has an active site complementary to substrate starch, and is able to hydrolyze the α -1,4 glycosidic bonds to produce maltose, a disaccharide of 2 α -glucose monomers joined together.

2. [effect of temp on enz = 1 mark]

- Amylase is a protein molecule with a specific 3D configuration maintained by intramolecular and intermolecular bonds between amino acid R-groups (eg. H-bonds, ionic bonds, disulfide bonds, hydrophobic interactions).
- Increase in temp and the increase in kinetic energy causes enzyme molecule to vibrate vigorously/ undergo thermal agitation such that intermolecular bonds are disrupted $\rightarrow$ enzyme loses specific 3D configuration $\rightarrow$ active site no longer able to accommodate substrate starch $\rightarrow$ less E-S complexes formed per unit time $\rightarrow$ enzyme is said to be denatured.

3. [Identification of variables & methods of measuring the dependent variable = 1 mark]

- Dependent variable: Rate of reaction measured by the largest rise in temperature of amylase-starch reaction over a 5-min period, in $^{\circ}\text{C min}^{-1}$.
- Independent variable: concentration of amylase / %
- Other variables to be kept constant:
 - o pH by adding citrate phosphate buffer into reaction mixture/ w.t.e.
 - o temp at which amylase-starch reaction is occurring by conducting experiment in automated thermostatically water bath maintained at 37°C .
- Reasons for keeping variables constant:
 - o pH and temperature must be kept constant because rate of enzyme activity is affected by changes in pH and temp.

Precaution:

4. [Precaution & safety = 1 mark – any 2]

- Mercury is toxic. Exercise caution in handling thermometer so as to prevent breakage.
- Glassware is fragile, handle with care to prevent breakage.
- Buffer is a skin irritant, wear gloves and goggles when handling.

Procedure**5. [specify appropriately at least 5 different concentrations of amylase at regular intervals in % = 1 mark]****Step 1:**

- Dilution table required.
- Dilute the 10% amylase solution using distilled water/ buffer shown in the dilution table.

Dilution Table

Boiling tubes	Volume of 10% amylase solution/ cm^3	Volume of distilled water / cm^3	Final vol/ cm^3	Final conc of amylase solution/ %
A	10.0	0.0	10.0	10.0
B	8.0	2.0	10.0	8.0
C	6.0	4.0	10.0	6.0
D	4.0	6.0	10.0	4.0
E	2.0	8.0	10.0	2.0
Control	0.0	10.0	10.0	0.0

No need to indicate control in table, but must appear in text

6. [acclimatisation timing]**Step 2:**

- Set up a water bath maintained at 37°C and immerse boiling tube A containing 3 cm^3 of 10% amylase solution into it.
- Immerse a test tube containing 3 cm^3 of 1.0 moldm^{-3} starch solution into the same water bath.
- Allow both amylase and starch solutions to acclimatize to the temperature for 2-10 minutes before adding the starch solution into boiling tube A.

7. [measurement of increase in temp = 1 mark]**Step 3:**

- Upon adding the starch solution into amylase solution, start the stopwatch and record the initial temperature, T_0 .
- Record temperature readings every 30 seconds (T_x) until 3 minutes is over.
- Calculate the largest increase in temperature ($T_x - T_0$) over the stated period of time.
- Repeat the experiment with boiling tubes B, C, D, and E using the same volume of starch solution.
- Repeat the experiment using equal volumes of M1 and M2 to replace amylase solution, and record the increase in temperature.

8. [collection & processing of data = 4 marks]**Step 4:**

- a) Record raw data in Table 1 (including correct table headings + formulas). **[1m]**
- b) Calculate the rate of increase in temperature in $^{\circ}\text{C min}^{-1}$ by taking the average of a least 3 readings then divided by 3mins **[1m]**
- c) Plot rate of temp increase under different amylase concentration on Graph 1 **[1m]**
- d) Use graph to find concentration of active amylase present in M1 and M2 **[1m]**

9. [replicates and repeats = 1 mark]**Step 5:**

- Repeat Steps 2-3 twice obtain two further readings.
- Repeat the entire experiment once.

10. [control with distilled water = 1 mark]

- same setup as the experiment except that the 3cm^3 amylase solution has been replaced by distilled water.

Amylase Concentr ation /%	Temperature/°C																							
	0s (T ₀)			30s (T ₁)			60s (T ₂)			90s (T ₃)			120s (T ₄)			150s (T ₅)			180s (T ₆)			Average largest increase in temperature (T _A)	Rate of increase of temp / °C min ⁻¹ (T _A /3)	
	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	1	2	3	(T _x -T ₀)/3		
10.0																								
8.0																								
6.0																								
4.0																								
2.0																								
0.0																								
M1																								
M2																								

Formulas must be shown in table heading.

Table 1

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b), etc., as indicated in the question.

5

(a) Explain the normal functions of stem cells in a living organism. [6]

1. *undifferentiated, unspecialised cells with no specific tissue structures;;*
2. *unlimited self-renewal capacity;*
due to high level of telomerase actⁿ ;
capable of undergoing con't division / mitosis;
to replace damaged or worn out tissues;
3. *totipotent;*
zygotic SC;
4. *able to dy/dx into all cell types including extra-embryonic tiss / placenta;*
® extracellular mem
5. *pluripotent;*
ESCs found in inner cell mass of embryo / blastocyst;
6. *able to dy/dx into any cell type except extra-embryonic tiss / placenta;;*
7. *into the 3 dermal layers: ectoderm, mesoderm, endoderm;;*
8. *multipotent;*
ASC e.g. HSCs, found in the bone marrow;
9. *able to dy/dx into grp of cells with related f(x) / limited cell lineages / myeloid, lymphoid progenitor;*
e.g. give rise to RBCs, WBCs, lymphocytes;
10. *SCs dy/dx in response to internal & external signals;*
that influences gene expression;

(b) Discuss the advantages and limitations of plant cloning. [8]

Advantages (max. 4)

1. *rapid pdtn of many plants in short period of time;*
which are genetically identical;
® plants grow faster
2. *able to generate whole plant from a small portion of explant;*
facilitate growing of GM plants;
3. *prod hybrid offspring from sexually incompatible plants with desirable traits fr*

both plants;

using protoplast culture;

4. *able to prod disease free plants;*

by meristemic culture using rapidly proliferating bact / virus-free meristematic tiss as explant;

5. *to prod. plant spp that are difficult to propagate;*

e.g. difficult to pollinate / low seed pdtn / low seed germination;

6. *propagate plants limited by seasonal variations;*

possible to meet consumer demand for out-of-season plants;

® idea of independent of climate as plants are grown in lab. Plants are eventually transplanted into field.

Limitations (max 4)

7. *high risk of contamination / genetic uniformity increases susceptibility resulting in high losses;*

8. *exposure to high conc. of PGRs;*

plants tend to undergo mutation;

9. *plant cells may possess physiological memory;*

plants generated from non-juvenile cells may mature before they are fully grown;

10. *high cost due to need for sophisticated facilities & high labour input;*

e.g. sterile lab conditions / tiss culture room / special nutrient media / trained personnel / transplanting plantlets;

- (c) Comment on the statement "GM crops can make a relevant contribution to solving health problems in developing countries." [6]

1. *state appropriate example:*

Golden Rice;

GM to contain (crt 1 & psy gene that syn) β -carotene (precursor to vit A);

2. *state health problem GMO can solve:*

treat vit A deficiency in children in developing countries;

which may lead to vision impairment / blindness / other infectious diseases;

® poor eye-sight

3. *For*

- a. *rice is staple diet;*

can be consumed in larger amt / easier access to ensure adequate vit A intake as cf to o/r food sources;

- b. *rice is relatively inexpensive & non-seasonal;*

cf to o/r vit A sources like green leafy veg;

- c. *sustainable & cost-effective;*
cf to alternative like vit supplements;
- d. *may be biofortified / enhanced with o/r micronutrients;*
e.g. I₂ / Zn to reduce problem of malnutrition;

4. *Against*

- a. *β-carotene is lipid soluble;*
malnourished children w/o enough fat stores to absorb;
- b. *may not be the long term solution;*
cf to improved & varied diets, rising socioeconomic status;
poor infrastructure & govt policies may prevent access to rice;
- c. *may topple ecological balance;*
↑ levels of β-carotene may impact organism downstream of food chain;
- d. *rice eaten in large quantities could ➡ 'xcessive intake of vit A;*
➡ hypervitaminosis / vit A toxicity;

® GM salmon which is not a crop

® Bt corn which aims to increase yield which may improve food shortage but not targeted to solve health problems

[Total: 20]

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